

$S\Delta\phi$ -03 — Irreversibility as the Minimal Condition of Existence

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Working Paper — Sofience- $\Delta\phi$ Formalism Series
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Abstract

This document establishes a minimal ontological axiom: irreversibility constitutes the minimal condition for existence. Existence is defined not through substance, identity, cognition, or interpretation, but through transformations that cannot be fully reversed without loss of structural information. The purpose of this document is to fix a minimal invariant condition underlying dynamical existence independent of interpretive frameworks.

1 Definition — State

Let $\phi(t)$ denote the state of a system at time t . A state represents system configuration without assuming subjectivity, meaning, or ontology.

2 Definition — Transformation

A transformation between successive states is defined as:

$$\Delta\phi(t) = \phi(t + 1) - \phi(t)$$

3 Definition — Irreversibility

A transformation is irreversible if reconstruction of a prior state requires information not contained within the current state.

$\text{Recover}(\phi(t) \mid \phi(t + 1)) \neq \phi(t)$ without external restoration.

4 Postulate — Minimal Condition of Existence

A system is said to exist insofar as irreversible transformations persist.

$$\text{Existence} \Leftrightarrow \text{Persistent}(\Delta\phi_{\text{irreversible}})$$

5 Consequences

- Identity corresponds to accumulated irreversible transformations.
- History emerges from irreversible constraint on future transitions.
- Interpretation is secondary to irreversible change.

6 Principle

Existence is not defined by what a system is, but by what it can no longer fully undo.

7 Scope

This document defines only a minimal ontological condition. Applications to cognition, artificial intelligence, physics, or philosophy constitute extensions beyond this core.

8 Relation to Sofience- $\Delta\phi$ Formalism Series

This axiom is compatible with and foundational to the following works:

S $\Delta\phi$ -01 — Sofience- $\Delta\phi$ Minimal Core: Phase-Change Formalism of Existence (v1.1)

DOI: <https://doi.org/10.5281/zenodo.18730265>

S $\Delta\phi$ -02 — Subject as Interpretive Emergence: Extension of the Sofience- $\Delta\phi$ Minimal Core (v1.1)

DOI: <https://doi.org/10.5281/zenodo.18730296>

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